



Few-Shot Learning for Biodiversity Assessment

Semantic segmentation of habitat classes from UAV imagery with $K \in \{1, 5\}$ labelled examples per class

Group 7 SRP 2025/26

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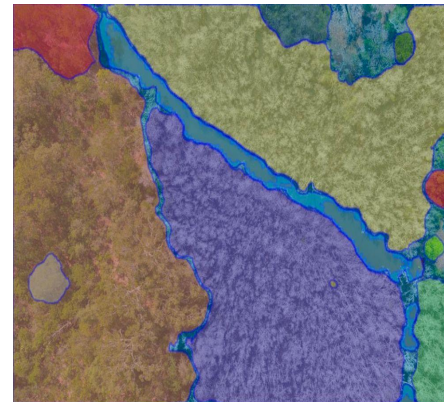
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Problem statement

Why this matters

- Biodiversity loss is accelerating due to climate change and human activity.
- Accurate, large-scale monitoring is critical for conservation and policy decisions.
- It is a global problem — different landscapes, climates, and vegetation across regions.
- Automated land-cover segmentation reduces the load on ecologists and policy makers.



Why UAV imagery

- High spatial resolution (0.1–0.3 m per pixel).
- Cost-effective and scalable to whole regions.
- Enables monitoring of vegetation types, species distribution, and habitat structure.

The technical bottleneck

- Pixel-level annotation requires domain experts — slow and expensive.
- Real biodiversity projects only have a handful of labelled images per class.
- Few-shot segmentation: learn from $K \in \{1, 5\}$ examples instead of thousands.

Research question & formal setting

Question

Can a model trained on base classes C_{base} learn to segment novel classes C_{novel} in unseen regions from K labelled support examples per class?

Few-shot semantic segmentation — formal setting

Support set $S = \{(x_i, y_i)\}_{i=1..K}$

K labelled (image, mask) pairs

Query image x_q

unlabelled, from same novel class

Predict $\hat{y}_q = f(x_q; S)$

segmentation mask for x_q

Objective maximise mIoU over C_{novel}

where $C_{\text{novel}} \cap C_{\text{base}} = \emptyset$

Related work — SAM & SAM2

SAM (Kirillov et al., 2023)

- ViT image encoder (ViT-H, 636M parameters) trained on SA-1B: 11M images, 1.1B masks.
- Prompt encoder supports points, boxes, masks, and text.
- Lightweight mask decoder produces class-agnostic segmentation — segments regions but does not assign semantic labels.

SAM2 (Ravi et al., 2024) — used in this work

- Hiera Vision Transformer — hierarchical, multi-scale features. We use Hiera-Small (~44M parameters).
- Trained on SA-V (video) + SA-1B, giving stronger spatial reasoning than the original SAM. ~6× faster on images.
- Empirical motivation: zero-shot oracle mIoU = 0.85 on LoveDA — the encoder already separates aerial structure well.

Related work — baselines we compare against

HRNet-W32 (Wang et al., 2020) — supervised baseline

- High-Resolution Network: maintains high-resolution features through every stage rather than downsampling then upsampling.
- Parallel multi-resolution branches with repeated cross-resolution fusion. W32 = 32-channel base width.
- Strong on fine boundaries — but requires full pixel labels for every scene.

DeepLabv2 + CBST (Zou et al., 2018) — Unsupervised Domain Adaptation

- DeepLabv2: ResNet-101 backbone with atrous (dilated) convolutions and ASPP for multi-scale context.
- CBST = Class-Balanced Self-Training: iteratively generates pseudo-labels on the unlabelled target with class-balanced confidence thresholds.
- Setup: train on labelled Rural domain, adapt to unlabelled Urban domain via pseudo-labels.

Related work — few shot and ecological indices



Wang et al. (2019) — PANet: Few-Shot Image Semantic Segmentation with Prototype Alignment

- Extends prototypical networks to semantic segmentation (the actual task we tackle).
- Uses masked average pooling to extract class-specific prototypes from support image-mask pairs.
- Achieves 48.1% / 55.7% mIoU on PASCAL-5ⁱ (1-shot / 5-shot) — sets the prototype-based baseline we build on.

Rouse et al. (1974) — Introduction of NDVI

- Original NDVI paper, developed for the NASA Earth Resources Technology Satellite (ERTS-1) Great Plains rangeland project.
- Defines $NDVI = (NIR - Red) / (NIR + Red)$; demonstrated strong correlation with green biomass and vegetation vigour.
- The most widely used vegetation index in remote sensing for over 50 years — established standard in ecology.

Wei et al. (2025) — Few-Shot UAV Vegetation Segmentation

- Most directly comparable prior work: combines UAV imagery, few-shot segmentation, and vegetation classes.
- Cross-matching + self-matching meta-learner over support-query features; SAM as a boundary refiner.
- Differs from our approach: meta-learning + matching rather than prototype matching; no ecological channels.

Our approach — overview

Gap addressed

No prior work combines SAM2 + prototype-based few-shot + NDVI/Shannon ecological channels + multi-country UAV evaluation in one pipeline.

Four components

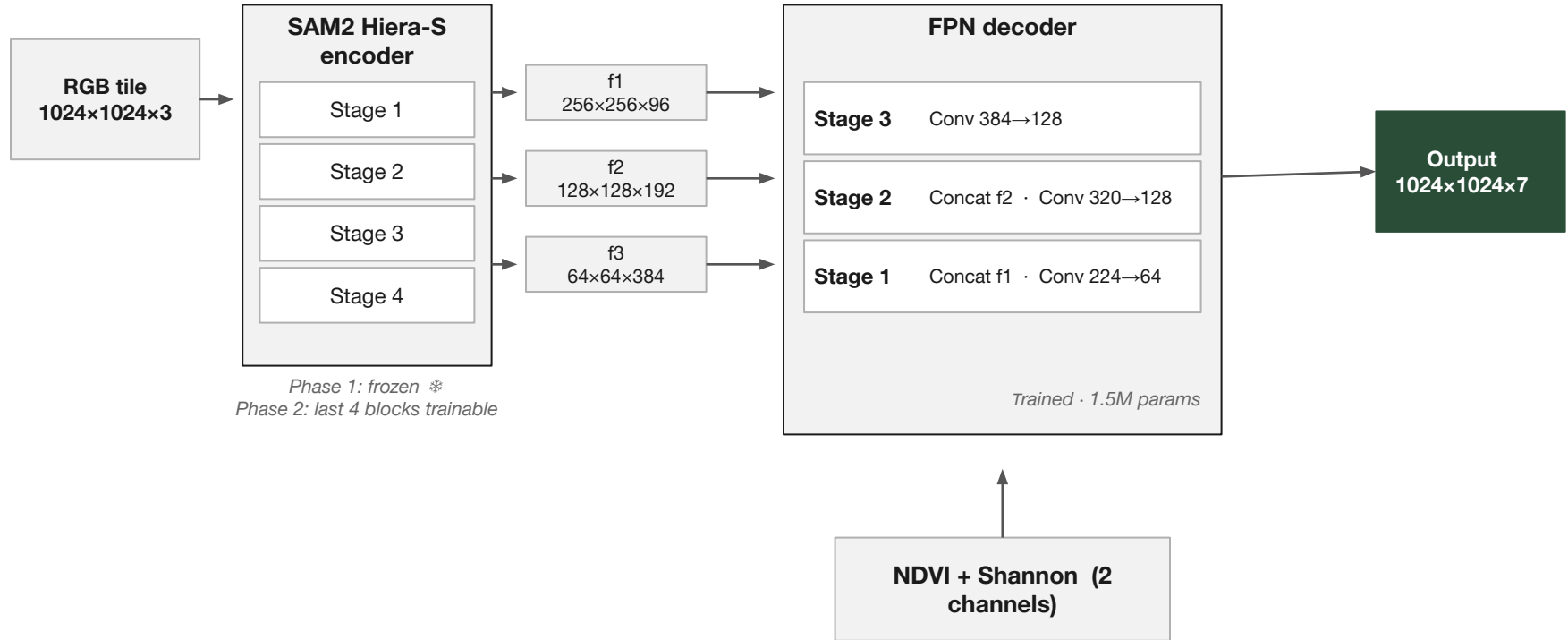
SAM2 Hiera-S encoder — Frozen in Phase 1; last 4 transformer blocks unfrozen in Phase 2. Provides multi-scale visual features.

FPN-style decoder — Trained head with skip connections that fuses encoder features at three resolutions back to per-pixel predictions.

Ecological channels (NDVI + Shannon) — NDVI captures vegetation health, Shannon captures habitat heterogeneity. Concatenated with encoder features at the decoder input. Encoder stays pure RGB.

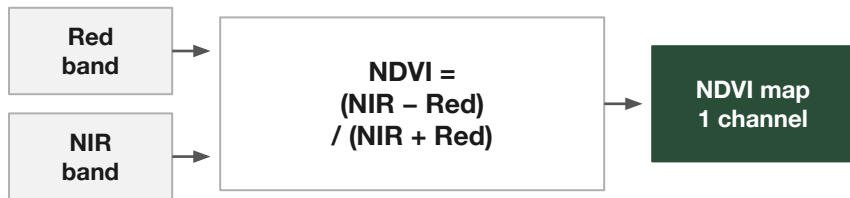
Prototype-based few-shot — Average encoder features over K support examples → class prototype. Query pixels assigned by cosine similarity.

Architecture – SAM2 encoder + FPN decoder + ecology fusion



Ecological channels — NDVI & Shannon diversity

NDVI — Normalised Difference Vegetation Index



Measures vegetation health. Healthy plants reflect NIR strongly → high NDVI. A field that looks green in RGB but is stressed, drought-affected, or degraded shows up clearly in NDVI. This is the signal ecologists use to assess vegetation vigour.

Shannon diversity index — spatial heterogeneity



$$H(p) = - \sum_i p_i \log p_i$$

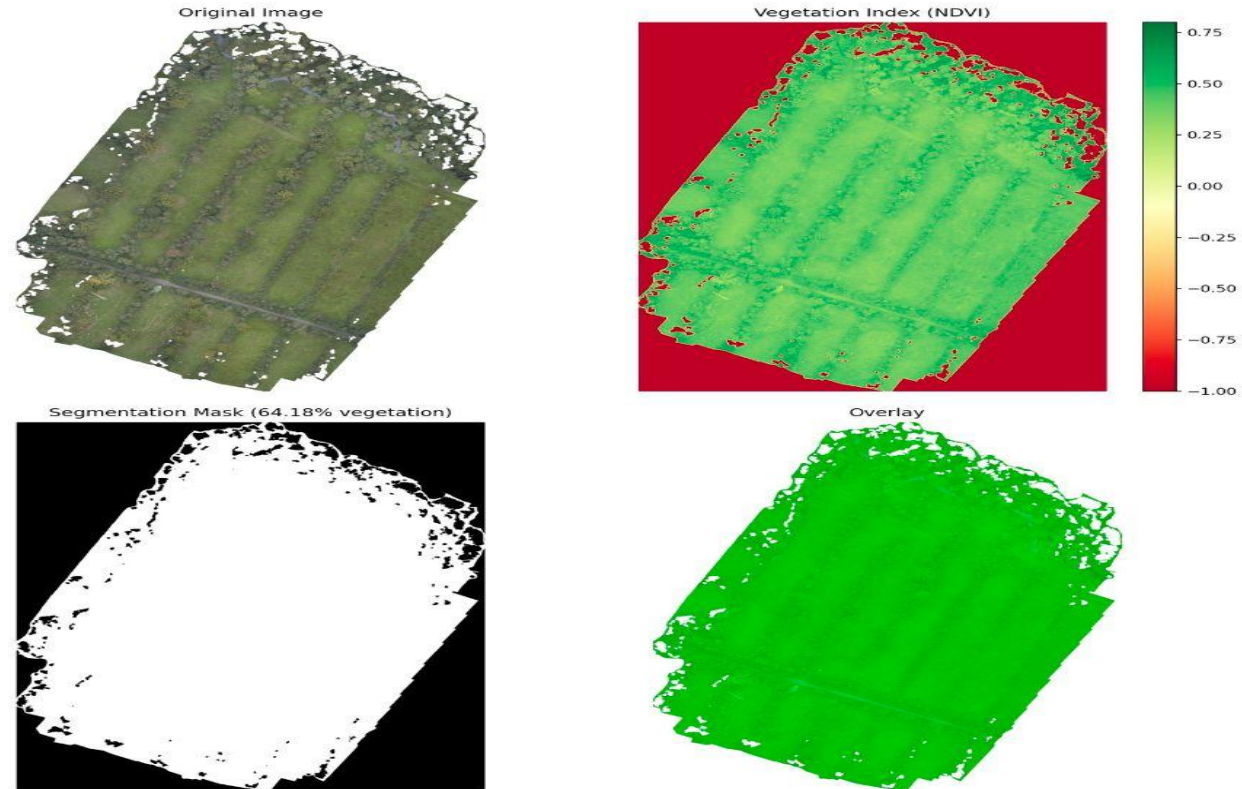
over k-bin NDVI histogram in window w

Measures habitat heterogeneity in a spatial patch. High H = mixed habitat (forest edges, mosaic landscapes — typically biodiversity-rich). Low H = uniform region (monoculture). A quantity ecologists already use in landscape ecology.

Where they enter the pipeline

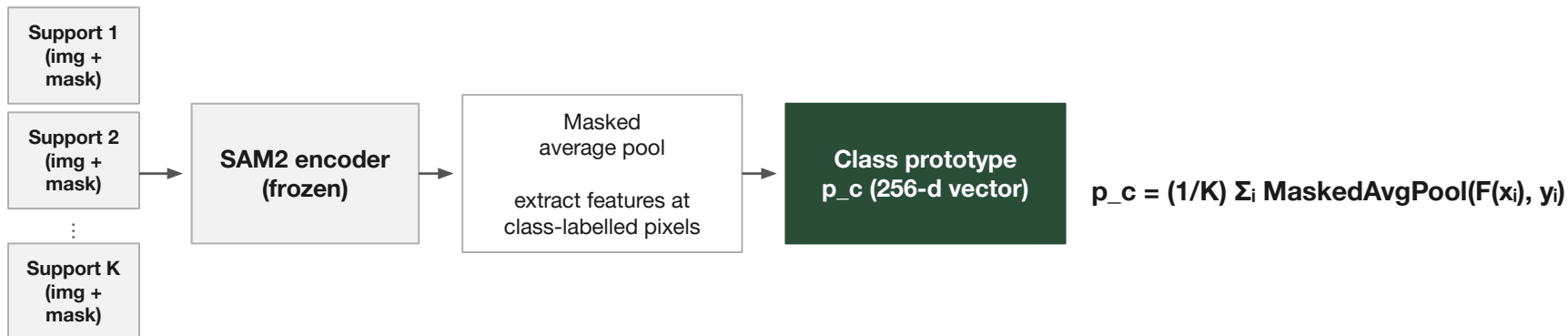
RGB only → SAM2 encoder (frozen). NDVI and Shannon are computed separately, downsampled to match each encoder feature's resolution, and concatenated with encoder features at the FPN decoder input. Encoder pretraining stays untouched.

NDVI Example



Few-shot mechanism — prototype matching

Step 1 — build a class prototype from K support examples



Step 2 — classify each query pixel by similarity to prototype



Overall pipeline — five staged experiments

	Stage / approach	Data setup
1	Supervised baseline <i>HRNet-W32, full labels</i>	LoveDA Urban+Rural
2	UDA baseline <i>DeepLabv2 + CBST, no target labels</i>	LoveDA Rural→Urban
3	SAM2 + FPN fine-tune <i>Phase 1 frozen / Phase 2 partial unfreeze</i>	LoveDA Urban+Rural
4a	Few-shot novel classes <i>Prototype matching from Phase 2 encoder</i>	LoveDA base→novel, $K \in \{1, 5\}$
4b	+ Ecology channels <i>NDVI + Shannon at decoder fusion</i>	Vaihingen
5	Cross-country transfer <i>Same pipeline, target country labels $K=5$</i>	LoveDA → EU UAV (DE first)

Datasets — LoveDA (public benchmark, used for validation)

Size: 5,987 tiles · 1024×1024 px · 0.3 m GSD

Classes: 7 — Background, Building, Road, Water, Barren, Forest, Agriculture

Domains: Urban (2,713) · Rural (3,274) — strong distributional shift

Sensor: RGB only · Google Earth · cities: Nanjing, Changzhou, Wuhan (CN)

Pixel distribution

Background 43% · Agriculture 24% · Forest 12% · Building 10% · Road 6% · Water 3% · Barren 1.7%

Heavy imbalance — Barren at 1.7% is the hardest class. Motivates inverse-frequency loss weighting.

Datasets — EU UAV & ISPRS Vaihingen

EU Multispectral UAV [Primary target]

- Countries: Portugal, Germany, UK, Austria.
- Sensors: RGB on all sites; NIR available on a subset.
- Habitats: grassland, forest edges, shrubland, agricultural fields, riparian zones.
- Labels: pixel-level on a subset — sparse, motivating the few-shot setting.

ISPRS Vaihingen [Supplementary]

- High-resolution aerial imagery from Vaihingen, Germany. Channels: RGB + Infrared + Digital Surface Model (DSM).
- Full pixel-level annotations across all tiles.
- Role: validate the NDVI/Shannon fusion under fully-supervised conditions before applying it to the sparse EU UAV labels.

Experiment plan

Stage	Method	Setup	Question answered
1. Supervised	HRNet-W32	LoveDA Urban+Rural, full labels	Best score with full supervision?
2. UDA	DeepLabv2 + CBST	Train Rural, adapt Urban (no labels)	Cost of removing target labels?
3. SAM2 fine-tune	Frozen ViT + FPN (Ph 1, 2)	LoveDA Urban+Rural	Does the foundation model help?
4a. Few-shot novel	Prototype matching	Base→novel split; $K \in \{1, 5\}$	Can prototypes segment unseen classes?
4b. + Ecology	Prototype + NDVI + Shannon	Same as 4a, decoder-side fusion	Do ecological channels help?
5. Cross-country	Same pipeline, K target labels	LoveDA → EU UAV (Germany first)	Does it transfer to new countries?

Stages 4a → 4b → 5 form the research contribution. Stages 1–3 are baselines and ablations.

Training details

	HRNet-W32	DeepLabv2 + CBST	SAM2 + FPN
Backbone	HRNetV2-W32	ResNet-50, OS=16	Hiera-S ViT (44M)
Input resolution	512×512 crop	512×512	1024×1024 full
Batch size	16	8	2
Iterations / epochs	15,000 iters	15,000 iters	20 epochs
Optimizer	SGD, mom=0.9	SGD, mom=0.9	Adam
Weight decay	1e-4	5e-4	default
Learning rate	0.01 (poly, p=0.9)	0.01	1e-4 (cosine)
Augmentation	Flip + Rot90 (p=0.75)	—	—
Training set	Urban + Rural	Rural (lbl), Urban (unlbl)	Urban + Rural
Trainable params	29M (all)	44M (all)	1.5M (decoder only)
Result (mIoU val)	0.58	0.351	0.4746 (Ph 1)

Stage 3 Phase 1 results — per-class mIoU on LoveDA



Class	HRNet	UDA	SAM2 ZS *	SAM2 FT
Background	0.53	0.32	0.70	0.5173
Building	0.56	0.55	0.78	0.5995
Road	0.55	0.44	0.88	0.4999
Water	0.68	0.59	0.97	0.6167
Barren	0.24	0.07	0.87	0.2285
Forest	0.42	0.23	0.78	0.3704
Agriculture	0.48	0.25	0.95	0.4901
mIoU	0.580	0.351	0.846	0.4746

Observations

Largest gains over UDA on minority classes:

Barren 0.07 → 0.23 (+0.16)

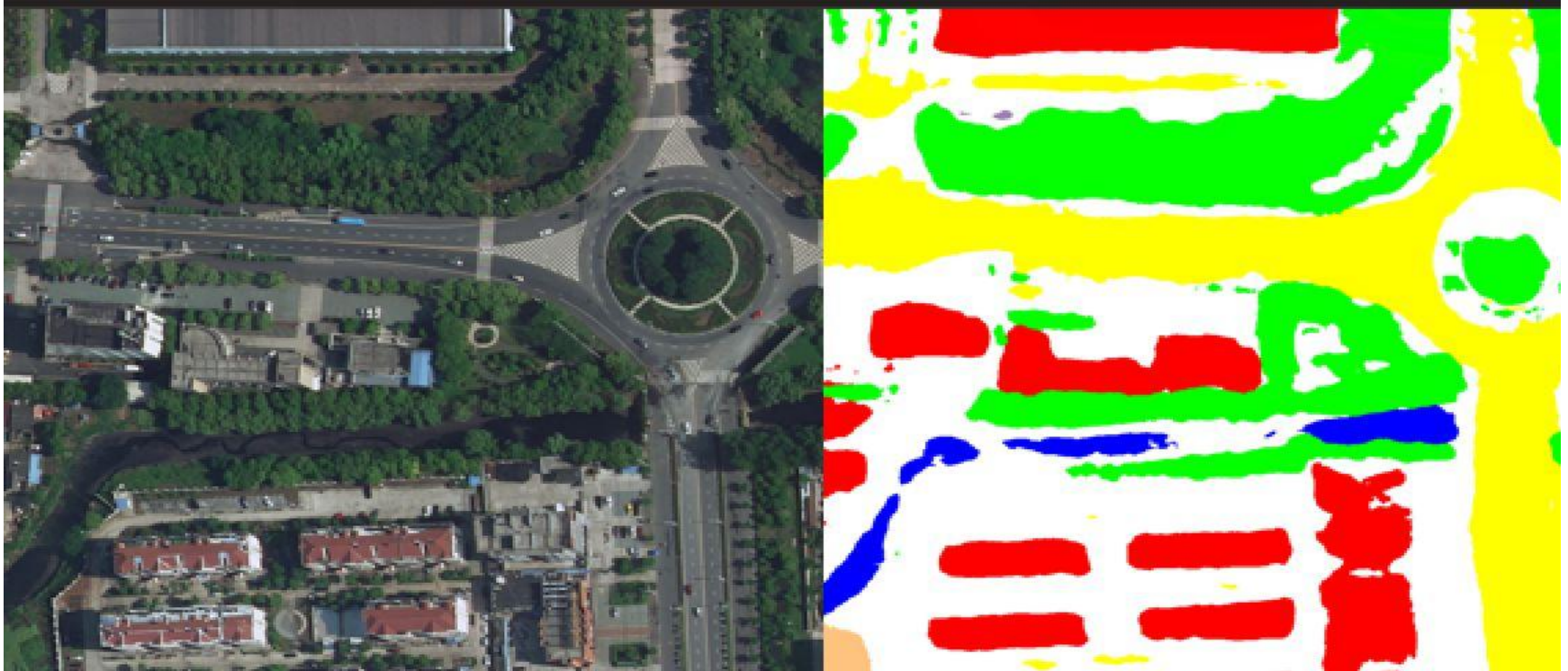
Agriculture 0.25 → 0.49 (+0.24)

Forest 0.23 → 0.37 (+0.14)

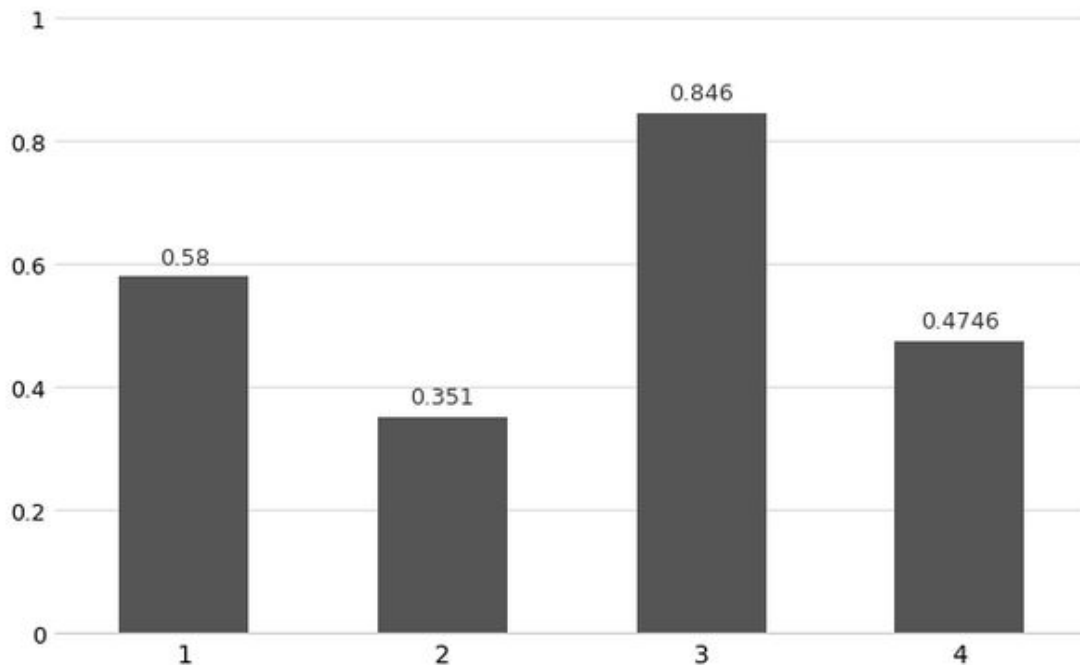
Gap to oracle (0.85) sits on fine spatial classes — motivates Phase 2 encoder unfreeze.

* SAM2 Zero-Shot uses ground-truth majority vote for class assignment — upper bound, not a usable model.

Supervised Result Visualized



Aggregate comparison



** oracle: ground-truth used to assign classes to SAM2's class-agnostic masks*

Interpretation

HRNet >> UDA:

HRNet (0.58) leads UDA (0.351) by 0.23
— CBST pseudo-labels fail to recover supervised performance under domain shift.

SAM2 zero-shot = upper bound:

0.85 with oracle labelling — encoder separates classes well.

SAM2 fine-tune:

SAM2 FT (0.47) trails HRNet (0.58) with frozen encoder — Phase 2 unfreeze targets 0.60–0.72 to close the gap.

Next steps & open questions

Immediate — Stage 3 Phase 2 encoder fine-tune

Unfreeze the last 4 Hiera blocks. LR $2e-5$ for encoder, $2e-4$ for decoder. 20 more epochs on the same cosine schedule. Open question: how much of the 0.85 oracle is reachable without overfitting?

Stages 4a & 4b — prototype-based few-shot on LoveDA, then with ecology

Split LoveDA classes into base / novel. Build class prototypes from $K \in \{1, 5\}$ support examples. 4a: prototypes from RGB features only. 4b: prototypes from RGB features + NDVI + Shannon (decoder-side fusion). Open question: do ecological channels improve over RGB alone, or is SAM2 sufficient?

Stage 5 — cross-country transfer to EU UAV

Validate the NDVI/Shannon fusion on ISPRS Vaihingen (full labels) before transferring to the EU UAV dataset — starting with Germany. Open question: does the prototype-based pipeline generalise across countries with only K labelled examples per class?